

# **Technology Stack (Architecture & Stack)**

## Project Design Phase-II Technology Stack (Architecture & Stack)

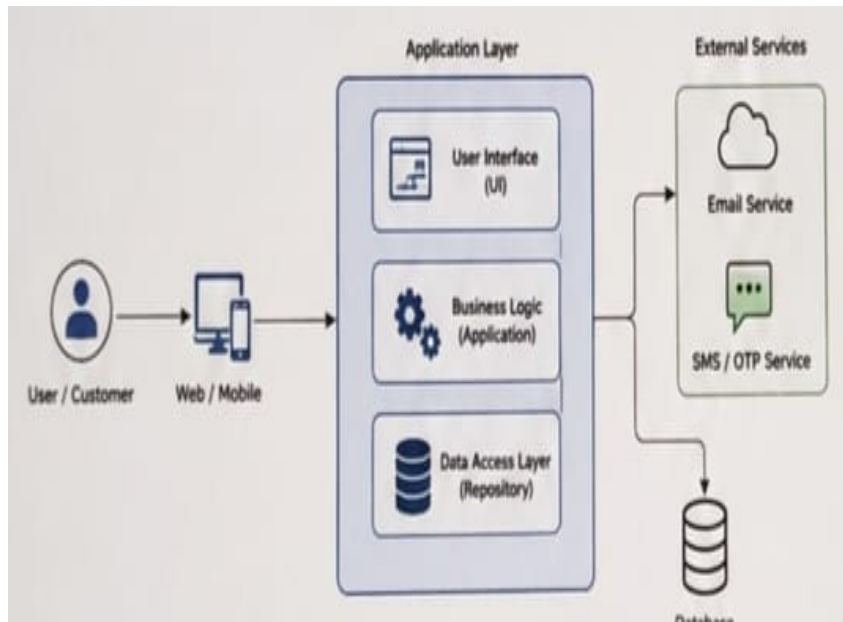
|               |                   |
|---------------|-------------------|
| Date          | 26 June 2026      |
| Team ID       |                   |
| Project Name  | Customer registry |
| Maximum Marks | 4 Marks           |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



### Guidelines:

1. Include all the processes (As an application logic / Technology Block)
2. Provide infrastructural demarcation (Local / Cloud)
3. Indicate external interfaces (third party APIs etc.)
4. Indicate Data Storage components / services
5. Indicate Interface to machine learning models (if applicable)

**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                                                | Technology                                         |
|------|---------------------------------|----------------------------------------------------------------------------|----------------------------------------------------|
| 1.   | User Interface                  | Web & Mobile interface for user registration, login and profile management | HTML, CSS, JavaScript / Angular Js / React Js etc. |
| 2.   | Application Logic-1             | Handle user registration, login, validation, and business rules            | Java / Spring Boot                                 |
| 3.   | Application Logic-2             | Manage emailSMS notifications and OTP verification                         | java                                               |
| 4.   | Application Logic-3             | RESTful APIs for communication between client and server                   | REST API                                           |
| 5.   | Database                        | Store customer and application data                                        | MySQL                                              |
| 6.   | Cloud Database                  | Cloud hosted database service                                              | AWS RDS(MYSQL)                                     |
| 7.   | File Storage                    | Store user uploaded documents and images                                   | AWS S3                                             |
| 8.   | External API-1                  | Email service API for sending verification emails                          | sendGrid/AWS SES                                   |
| 9.   | External API-2                  | SMS service API for OTP and alerts                                         | Twilio                                             |
| 10.  | Machine Learning Model          | (Optional) Fraud detection / user behavior analysis                        | Python(scikit-learn)                               |
| 11.  | Infrastructure (Server / Cloud) | Application deployment and cloud infrastructure Technology                 | AWS EC2, AWS S3, Docker, Kubernetes                |

**Table-2: Application Characteristics:**

| S.No | Characteristics        | Description                          | Technology  |
|------|------------------------|--------------------------------------|-------------|
| 1.   | Open-Source Frameworks | List the open-source frameworks used | Spring BOOt |

| S.No | Characteristics          | Description                                                                                         | Technology                   |
|------|--------------------------|-----------------------------------------------------------------------------------------------------|------------------------------|
| 2.   | Security Implementations | List all the security / access controls implemented (eg. Authentication, Authorization, Encryption) | JWT, BCrypt,HTTPS            |
| 3.   | Scalable Architecture    | Ability to handle growing number of users                                                           | Microservices, Load Balancer |
| 4.   | Availability             | System should be availability 24/7with minimal downtime                                             | AWS Cloud, Auto Scaling      |
| 5.   | Performance              | Fast response time for user requests                                                                | Caching (Redis), CDN         |

#### References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

# **Solution Requirements**

## Project Design Phase-II Solution Requirements

|               |                   |
|---------------|-------------------|
| Date          | 26 JUNE 2026      |
| Team ID       |                   |
| Project Name  | Customer Registry |
| Maximum Marks | 4 Marks           |

### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                       |
|--------|-------------------------------|------------------------------------------------------------------------------------------|
| FR-1   | User Registration             | Registration through Form<br>Registration through Gmail<br>Registration through LinkedIn |
| FR-2   | User Confirmation             | Confirmation via Email<br>Confirmation via OTP                                           |
| FR-3   | User Login                    | Login using Email & Password, Forgot Password                                            |
| FR-4   | Custer Profile Management     | Update Profile ,Change PassWord, Upload Profile Picture                                  |
| FR-5   | Customer Dashboard            | View Profile, Manage Account Details                                                     |
| FR-6   | Notification                  | Email Notifications, SMS Notification                                                    |

### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                              |
|--------|----------------------------|----------------------------------------------------------|
| NFR-1  | Usability                  | Easy- to-Use and User-Friendly interface.                |
| NFR-2  | Security                   | Authentication, AuthoriZation, HTTPS,Password Encryption |
| NFR-3  | Reliability                | The system should work reliably with minimal failures.   |
| NFR-4  | Performance                | Fast response time for registration and login requests.  |
| NFR-5  | Availability               | System should be available 24/7                          |
| NFR-6  | Scalability                | Should support a growing number of users.                |

# Data Flow Diagram & User Stories



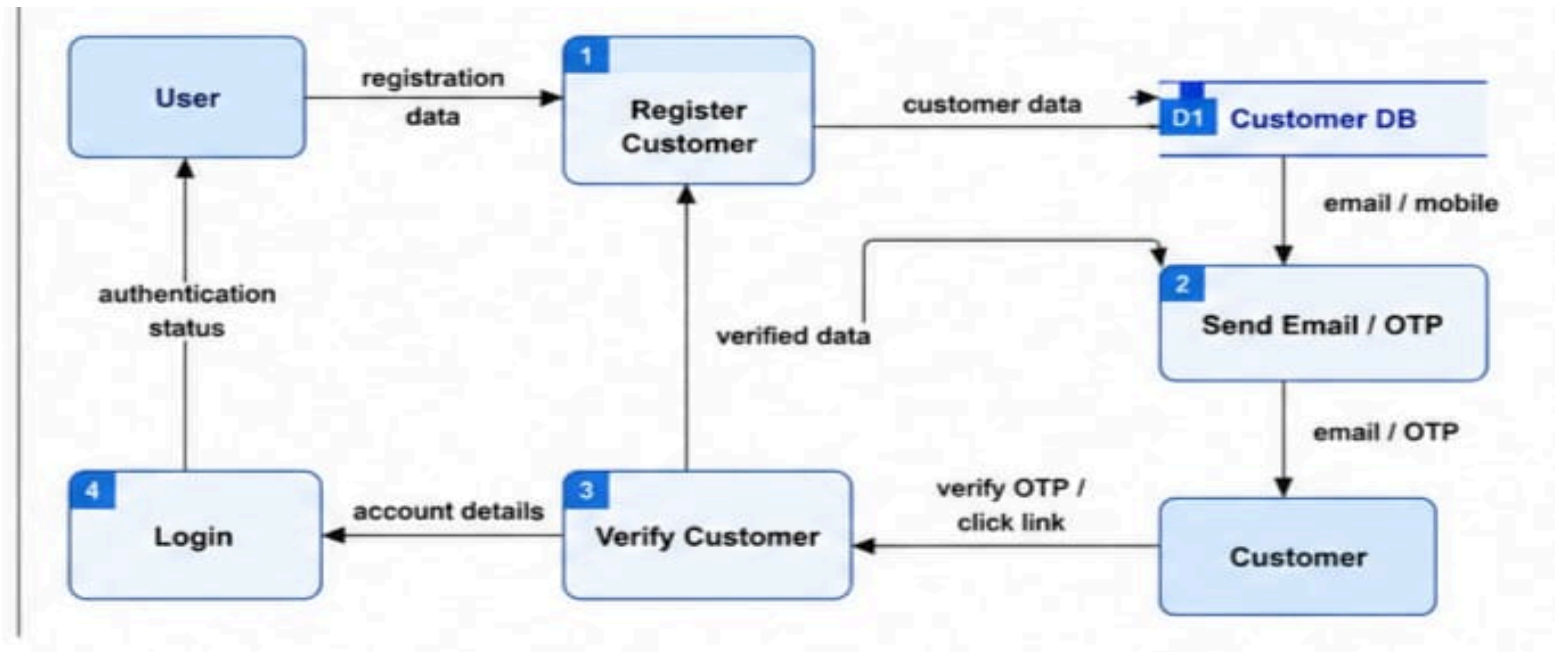
## Project Design Phase-II Data Flow Diagram & User Stories

|               |                   |
|---------------|-------------------|
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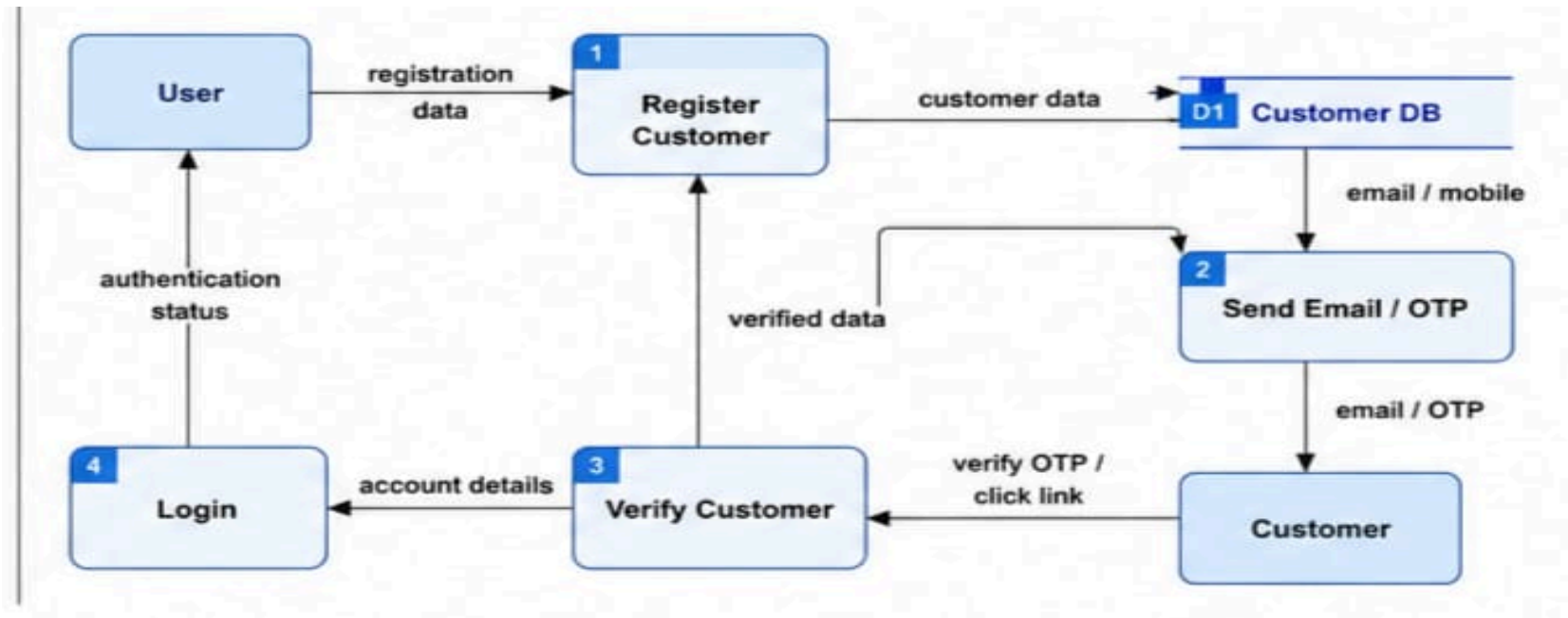
### Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)



### Example :DFD Level 0(Industry Standard)



### User Stories

Use the below template to list all the user stories for the product.

| User Type              | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                         | Acceptance criteria                              | Priority | Release  |
|------------------------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------|----------|
| Customer (Mobile user) | Registration                  | USN-1             | As a user, I can register for the application by entering my email, password, and confirming my password. | I can access my account / dashboard              | High     | Sprint-1 |
|                        |                               | USN-2             | As a user, I will receive confirmation email once I have registered for the application                   | I can receive confirmation email & click confirm | High     | Sprint-1 |

| User Type               | Functional Requirement (Epic) | User Story Number | User Story / Task                                                      | Acceptance criteria                                       | Priority | Release  |
|-------------------------|-------------------------------|-------------------|------------------------------------------------------------------------|-----------------------------------------------------------|----------|----------|
|                         |                               | USN-3             | As a user, I can register for the application through Facebook         | I can register & access the dashboard with Facebook Login | Low      | Sprint-2 |
|                         |                               | USN-4             | As a user, I can register for the application through Gmail            | I can register & access the dashboard with Gmail          | Medium   | Sprint-1 |
|                         | Login                         | USN-5             | As a user, I can log into the application by entering email & password | I can log in successfully                                 | High     | Sprint-1 |
|                         | Dashboard                     |                   | As a user, I can view my dashboard after successful login              | Dashboard displays my account information                 |          |          |
| Customer (Web user)     | Registration & Login          |                   | As a web user, I can register and login to the application             | I can register and login successfully                     |          |          |
| Customer Care Executive | Customer Management           |                   | As an executive, I can view and manage customer details                | I can search, view and update customer details            |          |          |
| Administrator           | User Management               |                   | As an admin, I can manage users and roles                              | I can create / update/deactivate users                    |          |          |
|                         |                               |                   |                                                                        |                                                           |          |          |
|                         |                               |                   |                                                                        |                                                           |          |          |
|                         |                               |                   |                                                                        |                                                           |          |          |
|                         |                               |                   |                                                                        |                                                           |          |          |